

$$x \frac{dx}{dt} + t = 1. \quad x \frac{dx}{dt} = 1 - t. \quad \int x \frac{dx}{dt} dt = \int x d(x(t)) = \frac{1}{2} x^2(t) + C.$$

$$\frac{1}{2} x^2 = t - \frac{t^2}{2} + C. \quad x^2 + t^2 - 2t = C.$$

$$x^2 + (t - 1)^2 = C, \text{ где } C > 0.$$

